

Sanyam Brar

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Software Engineering student based in **Annerley, Brisbane, Australia**, focused on systems programming, backend engineering, performance, and developer tooling.

Skills

Languages: Zig, C++, C, Kotlin, Go, Python, TypeScript, Java

Systems: Linux, POSIX, concurrency, networking, WebAssembly

Tools: Android, Git, SolidJS, PyTorch, libGDX

Projects

[zhtml](#)

Zig

High-throughput HTML parser and CSS selector engine with compact DOM storage, streaming APIs, lazy decoding, and conformance tests; exceeds **2 GB/s average parsing throughput** on its published benchmark suite.

[Reverb](#) [\[Demo\]](#) [\[F-Droid\]](#)

Android / Kotlin

Published rolling-audio recorder that keeps a rolling history across restarts, with streamed WAV export, playback/file management, live visualization, and hardware-aware recording controls.

[mnist-cnn](#) [\[Demo\]](#)

Zig / Python / WebAssembly

PyTorch-trained CNN with native Zig and browser WASM inference; embeds a **463 KiB model** into a self-contained browser build and reaches **99.43% MNIST accuracy**.

[Superintelligence 2077](#)

Java / libGDX

Team Java/libGDX game; built codex and death-state systems, persistent death markers, and cutscenes, alongside gameplay fixes, refactors, and automated tests.

Open Source

[btop](#) [\[writeup\]](#)

C++

Fixed broken lock ordering and related synchronization issues in btop, including data races, incorrect atomic memory ordering, shared-state guards, unsafe signal handling, and a memory leak.

[http.zig](#)

Zig

Contributed memory-leak fixes, reduced parser allocations, small hot-path optimizations, and build updates for newer Zig versions.

[Fiber](#)

Go

Fixed request binding in Fiber, including a nil-pointer panic in successful `Must` binding paths and related context handling.

Achievements & Leadership

- **1st place**, Creative Computing Society coding competition, 150 participants.
- **3rd place**, university competitive programming contest; **Top 5 / 400 teams**, overnight data privacy hackathon.
- **Core Member, GDSC & Creative Computing Society:** selected from 1,800 applicants; organized a 500+ participant technology event and taught DSA and Go workshops.